



Project Report

Daniyal Saeed

Muhammad Sajid

Tic Tac Toe Game

1. Introduction

This project implements a Tic Tac Toe game in 8086 assembly language, designed for execution on the emu8086 emulator. It features a command-line interface where two players alternate turns to mark positions on a 3x3 grid. The program manages game states, detects wins or draws, and outputs appropriate messages.

2. Objective

The objective of this project is to demonstrate the capabilities of 8086 assembly in creating an interactive game. This includes managing user inputs, updating game states, and providing feedback in real time.

3. Features

- Game Board Initialization: A 3x3 grid represented and manipulated via memory pointers.
- Player Interaction: Prompts players to input positions and validates the moves.
- Win Detection: Identifies row, column, or diagonal matches for victory.
- Draw Detection: Checks if the board is full without a winner.
- Customizable Messages: Displays start, win, and game-over messages.

4. Code Analysis

The program uses various assembly structures for efficient operation:

- Data Segment:

Stores game elements like the grid, player symbols, and status messages.

Example: ```asm`

```
game_draw db "_|_|_", 13, 10
           db "_|_|_", 13, 10
db "_|_|_", 13, 10, "$"
````
```

### - Stack Segment:

Allocates space for runtime operations.

```
``asm stack
segment dw
128 dup(?) ends
````
```

- Code Segment:

Initializes the game and manages the main loop.

Core subroutines include:

- ``set_game_pointer``: Initializes the grid.

- ``clear_screen``: Refreshes the display.
- ``print``: Outputs messages to the screen.
- ``main_loop``: Handles player turns and checks game status.

5. User Interface

The game operates entirely via the command line:

- Players are prompted to "TYPE A POSITION."
- The system updates the grid and checks for victory conditions after every turn.

6. Conclusion

This project demonstrates the use of assembly language in creating a functional and interactive game. It highlights concepts like memory management, stack operations, and subroutine calls. The Tic Tac Toe game not only provides a learning platform for assembly programming but also showcases its capability in implementing complex logic.